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Observer-specific orientation biases reveal distinct uncertainty-dependent modulation

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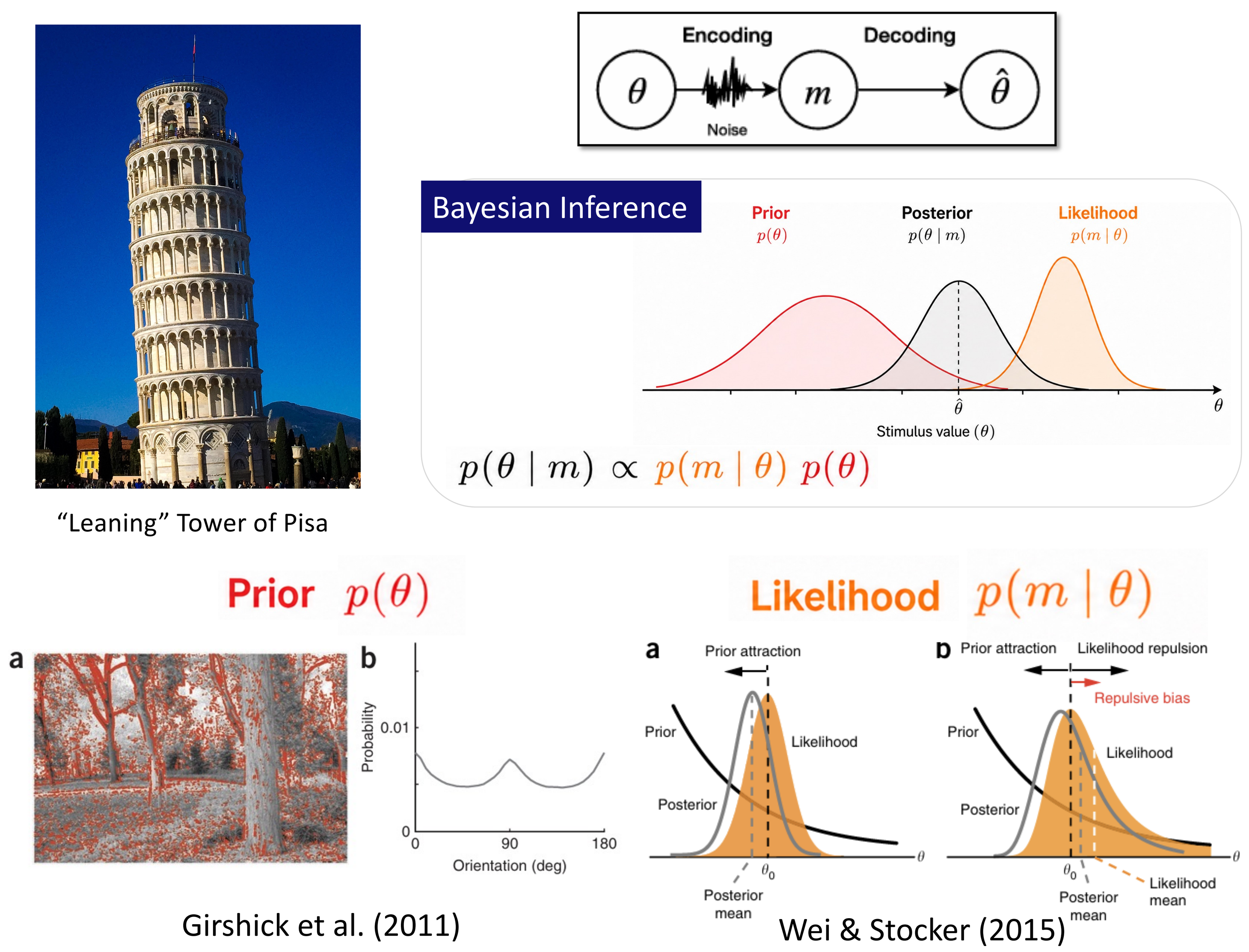
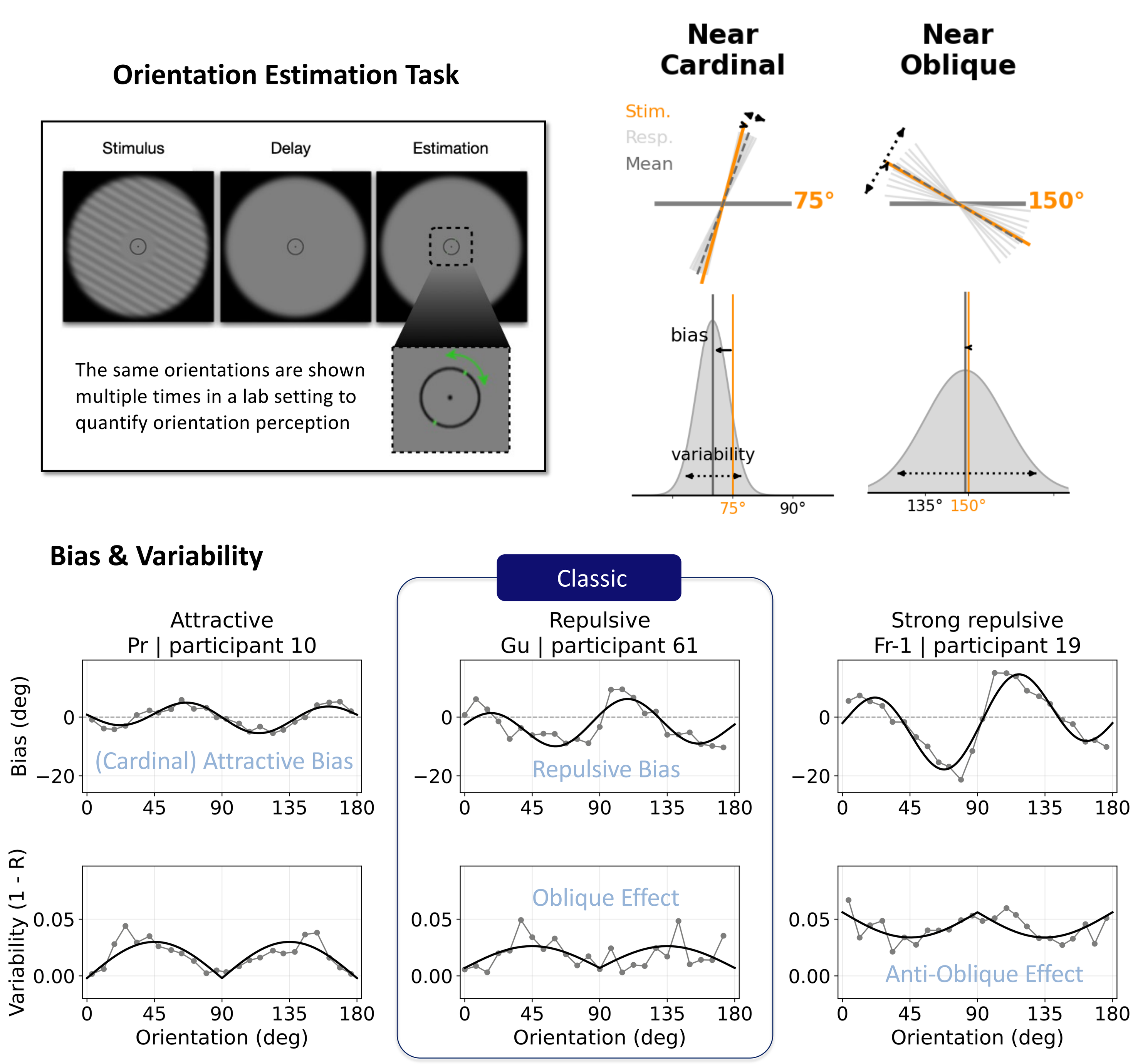
Background



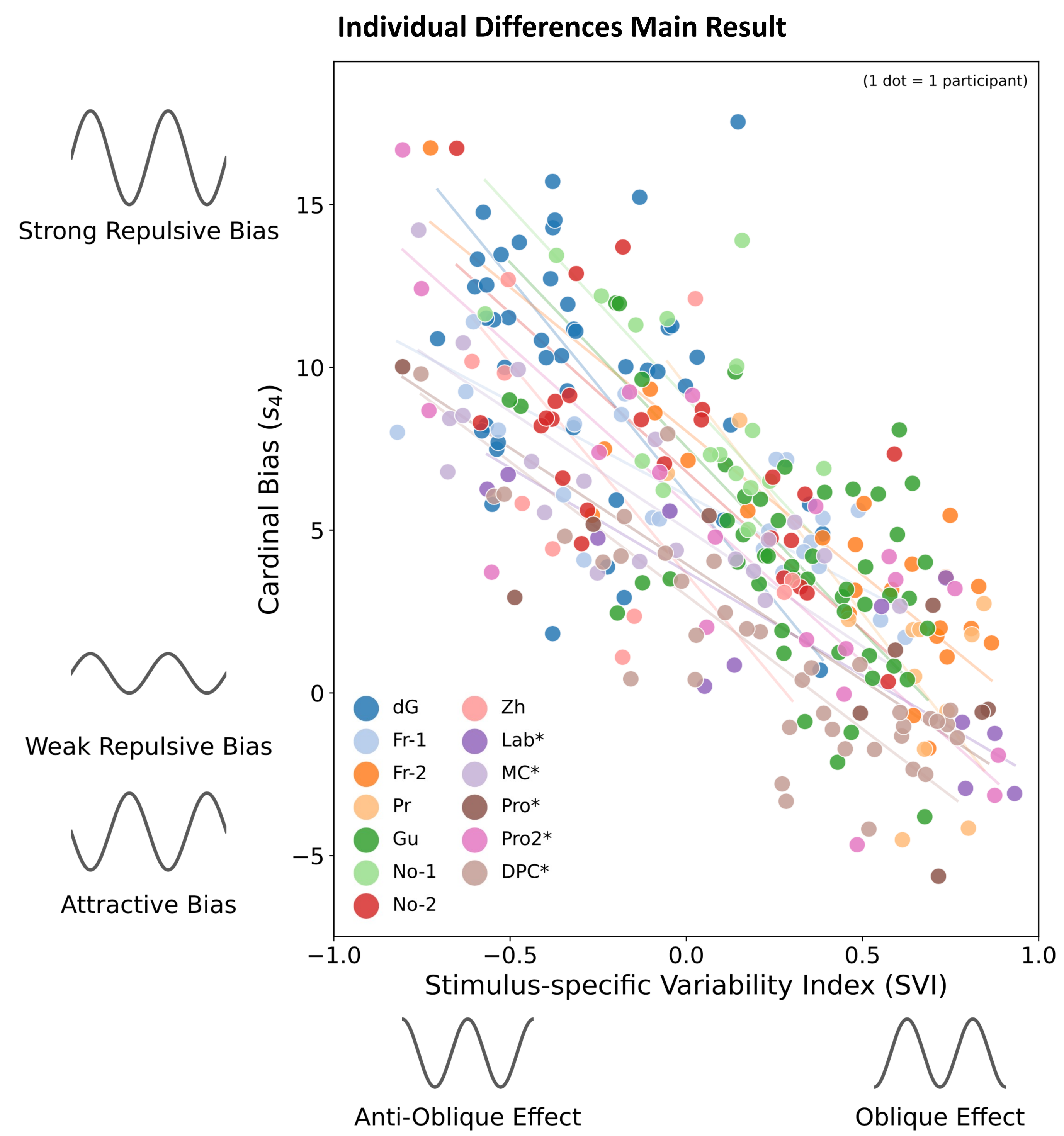
Orientation Estimation Task Analysis



- Bias: How estimates deviate from the true stimulus on average
- Variability: How widely the estimates are distributed around their average

Results

Result 1. **Individual differences** in bias and variability follow a **systematic** relationship



13 datasets/cohorts from published and in-lab studies

dG	de Gardelle et al. (2010), *Journal of Vision*
Fr-1, Fr-2	Fritsche et al. (2017), *Current Biology*, separate experiments
Pr	Pratte et al. (2017), *Journal of Experimental Psychology*
Gu	Gu et al. (2025), *Neuron*
No-1, No-2	Noel et al. (2021), *PLoS Biology*, ASD and TD cohorts
Zh	Zhang et al. (2025), *PNAS*

*** Lab, MC, Pro, Pro2, and DPC were collected within our lab**

❑ Y axis: Cardinal Bias (s4)

$$b(\theta) = \beta_0 + c_2 \cos 2\theta + s_2 \sin 2\theta + c_4 \cos 4\theta + \underline{s_4} \sin 4\theta$$

$s_4 < 0$: attractive | $s_4 > 0$: repulsive

❑ X axis: Stimulus-specific Variability Index (SVI)

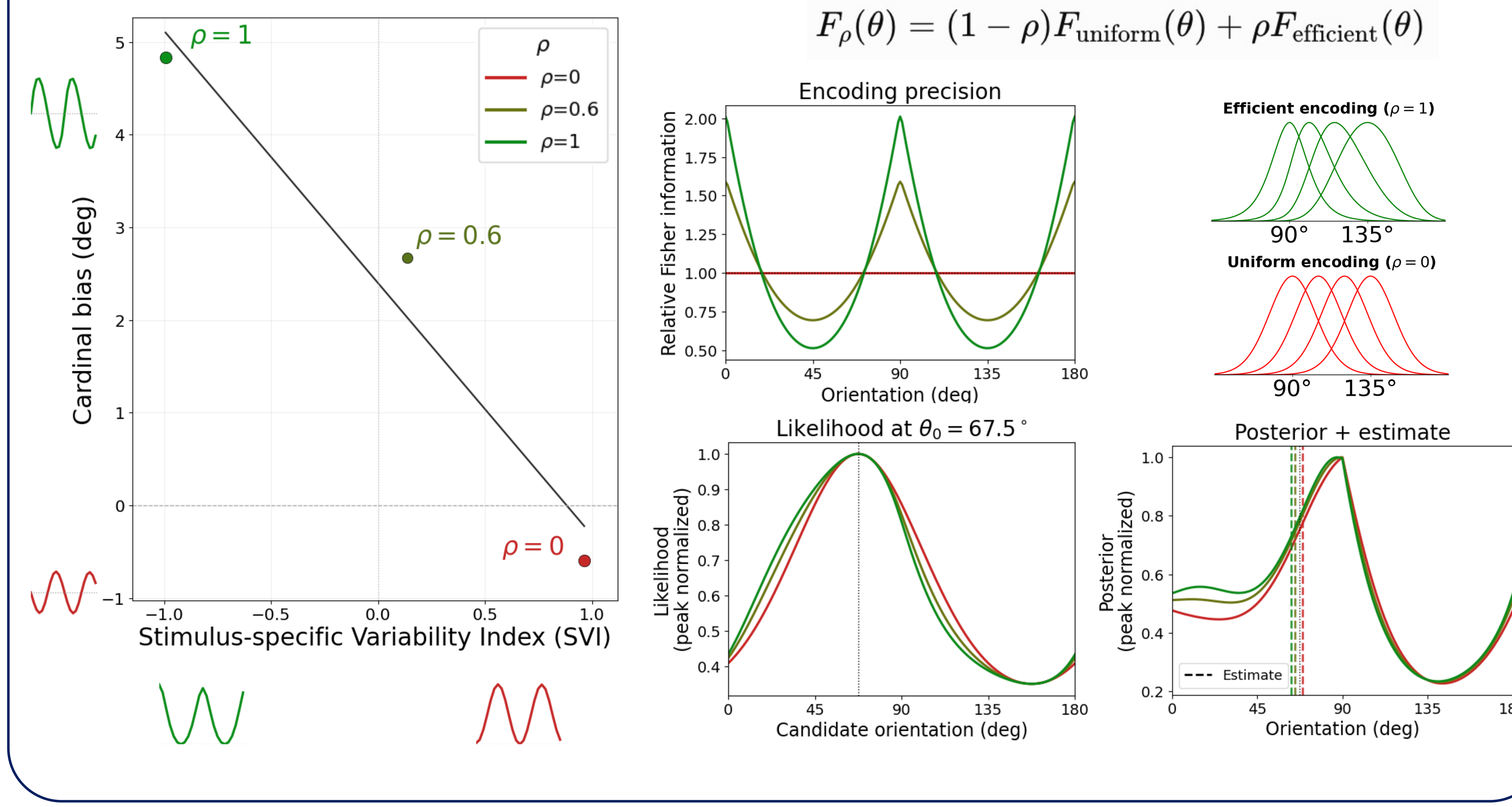
$$R(\theta) = \left| \frac{1}{N} \sum_j e^{i2e_j} \right|, \qquad V(\theta) = 1 - R(\theta)$$

$$\mathrm{SVI} = r\,[V(\theta), |\sin 2\theta|]$$

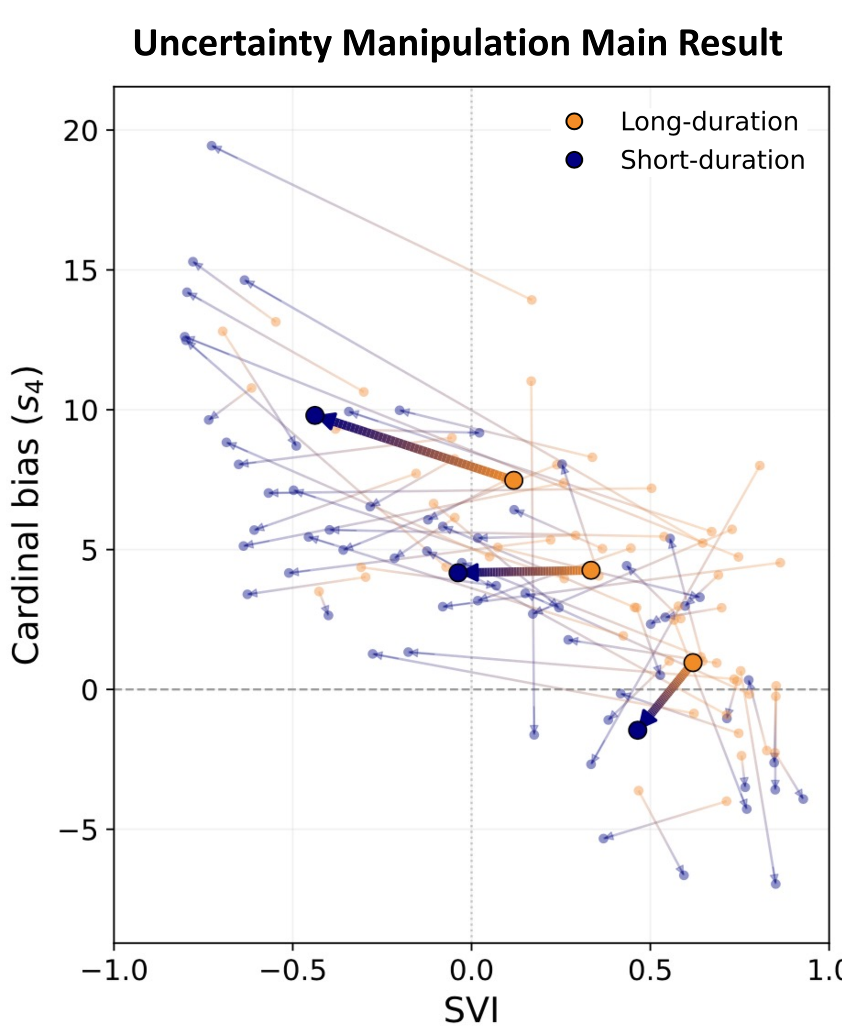
$SVI < 0$: anti-oblique | $SVI > 0$: oblique

Simulation

$$F_\rho(\theta) = (1-\rho) F_{\text{uniform}}(\theta) + \rho F_{\text{efficient}}(\theta)$$



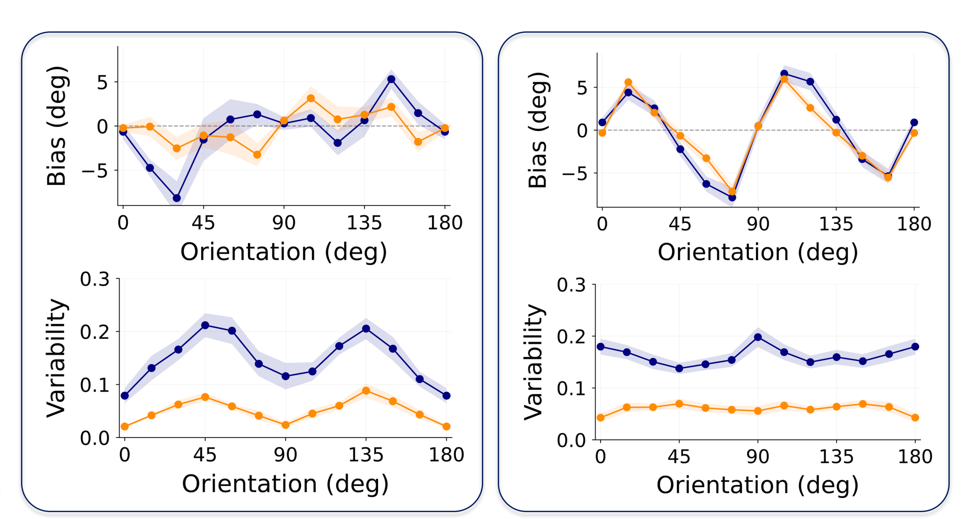
Result 2. **Sensory uncertainty** modulates orientation bias in an **observer-specific** manner



4 datasets with within-observer duration manipulation

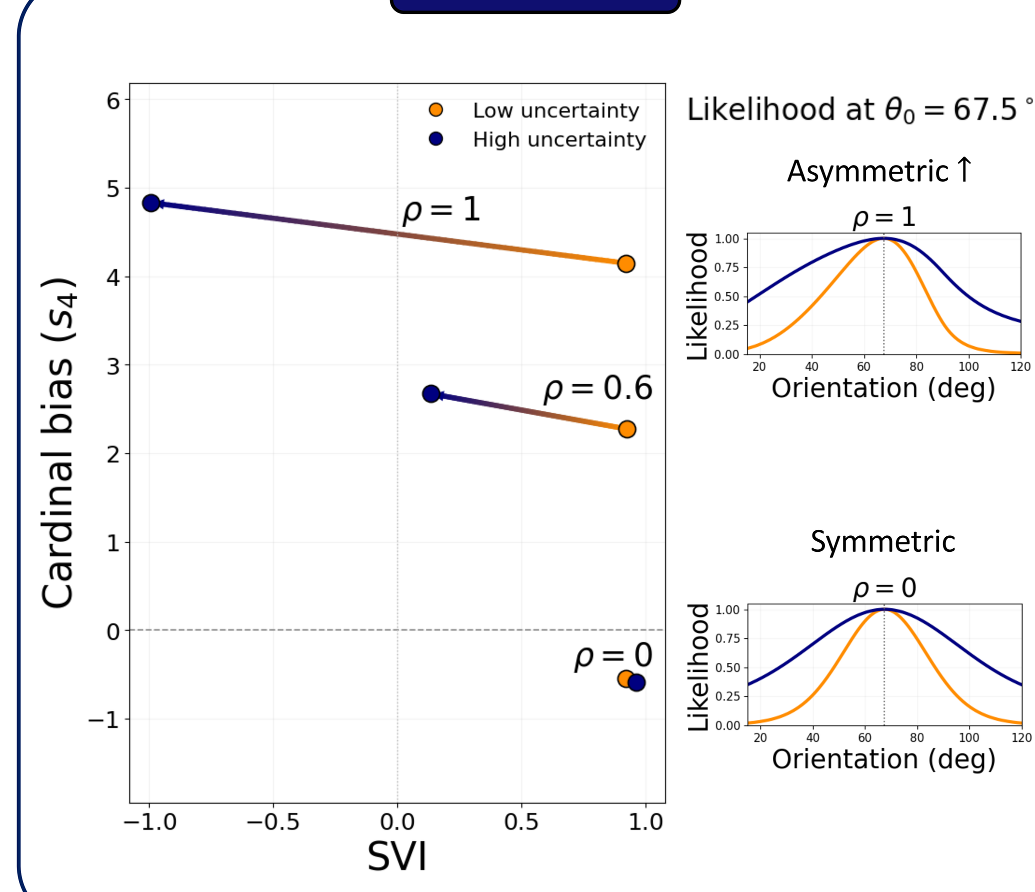
	Dataset		N	Target duration (ms)
1	Online prolific	Pro	N=10	34 / 167 / 500
2	Offline lab exp	Lab	N=12	50 / 500
3	Graduate Class	MC	N=20	50 / 500
4	Online prolific 2nd	Pro2	N=20	75 / 500

Attraction Group & Repulsion Group Profile Changes: Short vs Long



Attraction Group (n=13)   Repulsion Group (n=49)

Simulation



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